

UNIT 4

Ghana's Environmental Protection Bureau
&
Environmental Impact Assessment

Course Outline

- Ghana's Environmental Protection Bureau
- EIA
- Some Principles of Environmental laws.
- Sustainable Development Goals

Ghana's Environmental Protection Bureau, EIA and Some Principles of Environmental laws

- With the massive pollution problems mankind is facing, each country in addition to the provision of the concerted efforts globally, is supposed to address its own **local environmental challenges**.
- It is in line with these directives that Ghana has established its own **Environmental Protection Bureau (EPA)** which supervises the implementation of policies and regulations to protect the local Environment.
- One of such key regulation is the requirement for all developmental projects to undertake an **Environmental Impact Assessment** before project implementation.

Ghana's Environmental Protection Agency

- In order to implement all these environmental regulations an Agency has to be established to develop the laws and regulations to manage that aspect.
- The **Environmental Protection Agency**, is an agency of Ministry of Environment, Science, Technology and Innovation, established by EPA Act 490 in **1994**.
- Prior to that it was just an advisory council called **Environmental Protection Council** established in **1974** after a world conference on Environment in Stockholm **1972**.

Functions of the EPA

- The agency is dedicated to improving, conserving and promoting the country's environment and striving for environmentally sustainable development with sound, efficient resource management, taking into account social and equity issues.
- It oversees the implementation of the National Environmental Policy.
- EPA Ghana's mission is to manage, protect and enhance the country's environment and seek common solutions to global environmental problems.

Functions of the EPA

- Its mission is to be achieved through an integrated environmental planning and management system with broad public participation, efficient implementation of appropriate programmes and technical services, advice on environmental problems and effective, consistent enforcement of environmental **laws** and regulations.
- EPA Ghana is a regulatory body and a catalyst for change to sound environmental stewardship.

Some key environmental laws, policies and regulations in Ghana

- Ghana is still in the process of developing laws to protect the environment in addition to several environment laws dating back as 1892 on the Towns Ordinance which ensures proper planning of towns
- LI 1999 for the mandatory requirement for Environmental Impact Assessment for Development projects
- Public Nuisance Decree Act 29 of 1960 Prevent indiscriminate disposal of waste and noise disturbance.
- Ghana Investment code of 1988 which mandated all investors to examine the impact of their activities on the environment
- There are guidelines for the various levels of sound / noise for different places
- Guidelines for effluent discharges from industries
- Guidelines for ambient air quality
- Environmental sanitation policy of 2010. This needs to be revised.
- Water policy of 2007 also needs to be revised

Principles of Environmental Laws

- Several principles guides the development of these laws.
- **Precautionary**: This is done when the impact is not known but suspected
- **Prevention**: Impacts are known and the laws are made to prevent them. Most environmental laws are based on this principle
- **Polluter pays**: Ensures that industries pay for impact of the harmful effects of their products through life cycle of the product
- **Integration**: Integrates environmental concerns in all decisions of corporate bodies and activities
- **Public participation**: Incorporates the views of the public as well.
- **Sustainable development**: This is an approach to economic planning to foster development whiles preserving the environmental quality for present and future generations. This has been difficult to do because of the multipurpose nature of resources in question. Key sustainable laws are in ecotourism

Levels of Environmental Laws

- It exists at **three** main levels
 - **International** declarations, conventions and treaties
 - **National** laws are generated by agencies charged by Governments with the protection of the environment
 - **Local** laws such as the sanitation bye laws are developed by Assemblies

Incentives are also provided by some of these national laws to encourage the compliance of the laws. Such as pollution taxes for the use of clean technologies. Eg. Carbon tax

Types of Environmental laws

- Command and control
 - Identification of harmful effects
 - Imposition of specific conditions of standards for the control
 - Imposed conditions for failing to apply
- Environmental assessments mandates
 - Identification of a level of threshold potential impact which will require an EIA report
 - Establishment of specific goals for the assessment
 - Setting of requirements to decide whether to proceed with the action or not

Environmental Assessment

- There are several procedures for assessing environmental quality
- These are:
 - Environmental Impact Assessment (EIA)
 - Strategic Environmental Assessment (SEA)
 - Environmental Risk Assessment
 - Environmental audit etc
- We shall focus on EIA which now has social impact assessment as a key component

Environmental Impact Assessment

- EIA is a process of assessing **environmental impact of a development project** to improve decision making on its implementation

Brief History (Before the concept of EIA)

- Project evaluation – formerly prior to the execution of a major project there used to be **project evaluation**. This was based on technical studies valued in monetary terms (cost-benefit analysis).
- Rough evaluation – full of shortcomings and failures
- Efforts to develop new evaluation approach led to EIA. In this, Technical, Financial, Economic and Environmental aspects were given their proper weight in the decision making process.

Environmental Impact Assessment

- 1960 in the USA to create environmental awareness
- 1962 - First evidence of local application of **insecticides** had far reaching ecological impacts
- Environmental aspects of projects became a requirement for **decision making**
- National Environmental Policy Act / Law (NEPA) published in 1970. **Environmental Impact Statements (EIS)** should be published for major actions significantly affecting quality of human environment.

Purpose of EIA

- It enables financier/ client to take **environmental issues** into account
- It seeks to **compare the various alternatives** which are available for any project. Each alternative will have economic cost and benefit as well as environmental impact. Adverse impacts may be reduced at higher project cost whilst environmental benefits may be enhanced at environmental cost.
- It attempts to weigh the **environmental** effects on a common basis with **economic cost and benefit** in the overall project evaluation
- It **identifies and forecasts** the possible positive and negative impacts to the environment resulting from a proposed project.
- It helps to **avoid costs and delays** in implementation.

Some common targets

EIA seeks to identify probable alternatives for project with little or no adverse impacts on the community or environment.

EIS (Environmental Impact Statement) is the assessment report produced after the assessment exercise; it details both the beneficial and adverse impacts anticipated upon completion of a project and the probable impacts that would be experienced during the operational phase of the project.

To perform an EIA some **major parties** involved are :

- The **Proponent** or the Initiator e.g. Gov't agency, private personnel, company etc.
- **Decision makers** – Central or Local gov't agency, provincial official (EPA – Gh)
- **Review Commission** – EPA, EIA committee – management committee

It provides a formal mechanism for direct agency co-ordination to deal with concerns.

It measures the level of plan implementation and the degree of effectiveness of the above environmental protection provision.

The EIA Process

Includes

- Project description and the legal and administrative framework
- Scoping and screening which will look at the major impacts
- The description of the environment for the project
- Analysis of alternatives
- Impacts associated with the project with a check list of all things the project can have impact
- Mitigative measures in place

Environmental Assessment procedures

- A preliminary **form** is collected from the Ghana Environmental Protection Agency to determine the nature of the project whether an EIA is required.
- If it is needed the proponent seeks **consultants** to help in the development of the report which is assessed and approved by the EPA after public hearing.

Example of Environmental Quality Measurement

- Determine the Environmental quality index of a hair dressing salon at Ayeduase in a day if the number of people who come to the salon daily is 120.
- Each person uses 30 litres of water for her hair washing and styling.
- The pollution load in the water which is the impact is 40mg of Dissolved Organic matter per litre
- $Q=IFP$
- Q is the level of pollution of Environmental Quality index
- I is the impact per unit resource used ie organic matter dissolved per litre
- P is the size of the population, F is the amount of resource used per person
- Thus $Q= 40\text{mg/l} \times 30\text{L} \times 120=144,000$ in grams

Environment Impacts Associated With Various Engineering Disciplines

The impacts of the old engineering practices are known because they have been in existence for long:

- **Civil engineering** -changing of landscape, land degradation
- **Agricultural engineering** –land degradation, water pollution with pesticides, depletion of ground water resources
- **Mechanical, Materials, Chemical, Petroleum engineering** cause Noise pollution, fossil fuels depletion and air pollution
- **Computer engineering** impact details on man and the environment are not known apart from electromagnetic radiations.
- Impact of new disciplines/products will be better understood with time and also if a life cycle analysis is carried on the products developed.

Environmental Sustainability

- The United Nations Conference on Environment and Development in 1992 in Rio de Janeiro, Brazil
- Set out a roadmap for sustainable development.
- In a follow up conference in South Africa in 2002
- Why is there a lack of international progress in reducing poverty and protecting the environment? Discuss this.



The 3 Pillars of Sustainability

- **Environmental Protection** – Natural resource management, environmental management, pollution prevention
- **Economic Viability** – profit, cost savings, economic growth, research and development, employment
- **Social Equity** – standard of living, education, equal opportunity, community development, laws and ethics

Sustainable Development Goals

- In September 2015, the United Nations adopted the 2030 Agenda Sustainable Development Goals (SDGs) with 17 goals to replace the Millennium Development Goals (MDGs)
- The SDGs are to address the following dimensions of sustainable development:
 - Environment
 - Economic
 - Social

The time horizon is 2015-2030

The Sustainable Development Goals

- 1. No Poverty
- 2. Zero Hunger
- 3. Good Health And Well Being
- 4. Quality Education
- 5. Gender Equality
- 6. Clean Water And Sanitation
- 7. Affordable and Clean Energy
- 8. Decent Work and Economic Growth
- 9. Industry Innovation And Infrastructure
- 10. Reduced Inequality
- 11. Sustainable Cities and Communities



The Sustainable Development Goals

- 12. Responsible Consumption And Production
- 13. Climate Action
- 14. Life Below Water
- 15. Life On Land
- 16. Peace And Justice Strong Institutions
- 17. Partnership To Achieve The Goals



Incorporation of SDGs in National Development Agenda

- The National Development Planning Commission is the lead government Institution on Development Planning policy and strategy.
- The SDGs must be triggered down in our all our communities including all programs in our Universities to enhance the achievement of the goals

Facts for Water and sanitation

- 2.6 billion people have gained access to **improved drinking water** sources since 1990, but 663 million people are still without
- At least **1.8 billion** people globally use a source of drinking water that is **faecally contaminated**
- Between 1990 and 2015, the proportion of the global population using an improved drinking water source has increased from 76 per cent to 91 per cent
- But water scarcity affects more than **40 percent of the global population** and is projected to rise. Over 1.7 billion people are currently living in river basins where water use exceeds recharge
- **Floods** and other **water-related disasters** account for **70 per cent of all deaths related to natural disasters**

Facts continued

- 2.4 billion people lack access to basic sanitation services, such as toilets or latrines
- More than 80 percent of wastewater resulting from human activities is discharged into rivers or sea without any pollution removal
- Each day, nearly 1,000 children die due to preventable water and sanitation-related diarrhoeal diseases
- Hydropower is the most important and widely-used renewable source of energy and as of 2011, represented 16 percent of total electricity production worldwide
- Approximately 70 percent of all water abstracted from rivers, lakes and aquifers is used for irrigation

Goal 6 of the SDG

- By **2030**, achieve universal and equitable access to **safe and affordable drinking water** for all.
- By **2030**, achieve access to **adequate and equitable sanitation and hygiene** for all and **end open defecation**, paying special attention to the needs of women and girls and those in vulnerable situations.
- This goal affects the success of almost all the goals and hence the special attention to it.

8 Targets for SDG 6

- By 2030, achieve universal and equitable access to safe and affordable **drinking water** for all
- Achieve access to adequate and equitable **sanitation** and **hygiene** for all and end open defecation, paying special attention to the needs of women and girls and those in vulnerable situations
- **Improve water quality** by reducing pollution, eliminating dumping and minimizing release of hazardous chemicals and materials, halving the proportion of untreated wastewater and substantially increasing recycling and safe reuse globally
- Substantially increase **water-use efficiency** across all sectors and ensure sustainable withdrawals and supply of freshwater to address water scarcity and substantially reduce the number of people suffering from water scarcity
- Implement **integrated water resources management** at all levels, including transboundary cooperation as appropriate
- Protect and restore **water-related ecosystems**, including mountains, forests, wetlands, rivers, aquifers and lakes

SDG 6 Targets

- Expand **international cooperation and capacity-building support** to developing countries in water- and sanitation-related activities and programmes, including water harvesting, desalination, water-use efficiency, wastewater treatment, recycling and reuse technologies
- Support and strengthen the **participation of local communities** in improving water and sanitation management

Assignment I

80 welding shops produce 39 μg of Hg/m^3 from their foundry activities.

Each shop is confined to a space of 500m^3 .

What impact will this have on the health of the artisans after a long exposure?

Assignment 2

- Why are people not bothered about matters of the environment?

Reading Materials

- Visit Wikipedia for topics which are not too clear. Wiki has very good reference materials
- Living in the environment by George Tyler Miller 2005
- Introduction to environment and society by J. P. Evans 2012
- WHO Guidelines for Drinking water (current version)

The Epilogue

- The environment is a complex system requiring a proper understanding of how things are connected and how they impact each other. The environment in which we live have natural ecosystems that are self sustaining and self regulating. We can learn from nature on how it recycles matter in the ecosystem, manages its problems on water, air, and soil pollution and general interactions with other organisms in the environment.
- This concept of biomimicry which is the study of nature's most successful ideas over the past 3.5 million years, and adapt them for human use will go a long way to make our planet livable and healthy.